

IMPACTS OF FDI ON EXPORT: THE CASE OF ASIA COUNTRIES

Abstract

Foreign Direct Investment (FDI) is one of the important factors contributing to promoting the export of goods and services and expanding the international trade of each country. Along with increasing capital, FDI also transfers advanced technology to help improve production efficiency and improve product and service quality, thereby creating competitive advantages in the current export market. According to research results, when FDI increases by 1%, exports increase by 0.0625% for 47 countries in Asia region, 0.0789% for upper-middle-income countries and 0.105% for lower-middle-income countries. Besides the FDI factor, other macro factors such as GDP, exchange rate, inflation, population, trade openness, and HDI also have an impact on exports, and depending on the socio-economic development conditions, this impact may vary in countries. From the results of this study, the study proposes policies to attract FDI. At the same time, the study also proposes policies to improve the quality of goods and services; the qualifications and skills of the labor force; and policies to stabilize the socio-economy to create an effective investment environment, increase the competitive advantage of enterprises in the international market, and improve export performance.

Keywords: FDI, exports, Asia countries, SGMM.

1. Introduction

In the current era of international integration, foreign direct investment (FDI) has become a key factor promoting export efficiency and socio-economic development of each country. For Asian countries in the process of transitioning from an agricultural economy to an industrial-service economy, FDI plays an extremely important role in providing capital, technology, and production experience to help these countries improve production, expand export markets, and develop a strong economy. However, the impact of FDI on promoting exports in host countries is still a widely debated topic. Several studies have shown that foreign direct investment (FDI) can promote exports in a number of ways, including: (1) offering supplementary funding for export production; (2) promoting the transfer of technology; (3) enhancing and modernizing current product inventory; (4) developing the technical and management capabilities of employees in local companies; (5) ensuring access to international distribution networks for exporting goods in global markets; and (6) refining the export framework of the host nation (Sahoo & Dash, 2022). For a country, Sultan (2013) argues that FDI enhances a country's supply capacity by increasing factor supply in the form of additional capital inflows. Furthermore, it improves the competitiveness of national exports by increasing total factor productivity. At the industry level, UNCTAD (2005) demonstrated that in some countries, such as Singapore and China, FDI has contributed significantly to the transformation of the export structure. This transformation significantly increases the

technological content of exports and supports the development of supply capacity in industries (especially knowledge-based industries). Furthermore, the study shows that FDI not only upgrades technology but also contributes to structural improvements in the export sector, whereby industrial structure plays an important role in export performance in both the early and later stages of development. At the firm level, FDI inflows not only increase export volume but also increase the value of goods because most multinational corporations possess superior knowledge, experience, and advanced production technology and have closer links with the parent company's supply chain than domestic companies. Simultaneously, the spillover effects engendered by the production activities of multinational corporations can exert a beneficial influence on the productivity of domestic enterprises by enhancing operational and production processes as well as improving human capital. Consequently, individuals with superior education, equipped with advanced knowledge, skills, and competencies, in conjunction with innovative production methodologies, will generate higher-quality goods and services for consumers, thereby augmenting the export turnover of the host country receiving foreign direct investment (BAYAR & DIACONU, 2022).

However, the effectiveness of FDI in promoting exports depends largely on the scale, level of socio-economic development, and development context of the exporting and receiving countries. Therefore, there may be differences in the impact of FDI on exports between developing and developed countries; even when they are all developing countries, there are significant differences because the development level of each country is different (Sahoo & Dash, 2022).

Therefore, the question posed for this paper is: ***“Does foreign direct investment affect exports in Asian countries?”***. For this purpose, a sample of 47 Asian countries was collected with the analysis period of 1993–2022. In addition, if there is a relationship between these two factors, is there a difference between different economies classified into four groups according to the Organization for Economic Cooperation and Development (OECD): lower-middle income countries; middle-income countries; newly industrialized countries; OECD countries; and emerging countries? And what is the magnitude of the impact between these groups? However, due to the nature of the dataset, the countries will be divided into two groups: lower-middle-income countries (22 countries), and upper-middle-income countries (25 countries). The study will determine whether foreign direct investment inflows affect exports of countries in the Asian region and, if so, whether there are differences between countries with different economic backgrounds. Thereby, policies are proposed to improve the effectiveness of FDI in promoting exports, contributing to the economic development of countries in the Asian region.

The structure of the paper is as follows: Section 2 reviews the existing literature on the relationship between FDI and exports in different regions and countries; Section 3 details the data sources and research methods used in the paper; Section 4 presents the research results; and Section 5 summarizes the main findings and provides some policy recommendations.

2. Literature review

2.1. Relationship between FDI and exports

Foreign direct investment (FDI) is an important factor in socio-economic development and international trade. The impact of foreign direct investment (FDI) on the export of goods and services is a widely studied topic; however, with different economic contexts, foreign direct investment only affects a number of countries, with different levels of influence between countries (BAYAR & DIACONU, 2022).

Numerous studies have indicated a positive relationship between FDI and exports in both the short and long term. UNCTAD (2005) indicated that FDI is likely to affect export performance positively. FDI strongly contributes to the transformation of the composition of exports. Beside that, it showed that FDI inflows into Singapore or, more recently China, have helped to increase significantly the technological content of exports by supporting strongly the development of export supply capacity, including knowledge-based industries.

Another study using vector autoregression model (VAR), Bhatt (2013) indicated that 1% increase in FDI is associated with a 0.25% increase in exports over a one-year period. Beside that, Granger Causality Test indicated that there is a unilateral relationship between Exports and FDI and the direction is from FDI to Exports which means that FDI causes Exports. In addition, another study that using the VAR model but based on a group of bilateral data of China and 19 countries/regions in the period 1984–1998, LIU et al. (2001) showed that the increase in FDI inflows not only boosts production but also improves the quality of exports, playing a more important role than imports in generating exports. With VAR Model and based on annual time series data from 2001 to 2018 obtained from the Central Bank of Peru and the World Bank, Canchari et al. (2020) suggests that countries that pursue export promotion rather than import substitution policies will be more effective due to the impact of FDI, as FDI is one of the main drivers of export promotion and has a stronger impact than domestic capital investment

Moreover, with a Melitz-type theoretical model of firm heterogeneity combined with industry-level panel data from the Chinese manufacturing sector and measuring industry export quality by the unit value of industry exports, Anwar and Sun (2018) found that the presence of FDI inflows in China's manufacturing sector, especially the HMT sector, has led to a significant increase in the quality of China's exports.

In addition, using a panel regression model and annual data for BRICS countries and Türkiye, Egypt, Indonesia and Colombia for the period 1980 – 2020, BULUT and YAŞAR (2023) suggest that FDI has a positive impact on exports when exports tend to increase, but there is no impact at high export levels due to many reasons, such as direct investment to buy installed capacity instead of creating new capacity; production is carried out to meet domestic demand; privatization activities are common in developing countries; and a dynamic population as well as large markets also support this view.

Using the gravity model regression, Liu and Graham (1998) found that outward FDI tended to stimulate exports (mostly of capital goods and intermediate goods) without stimulating imports in equal magnitude in Taiwan and South Korea.

Using Vector Error Correction Model (VECM) with the period annual data from the World Bank online databas from 1981 to 2015 in Indonesia, Cora and Wen (2020) found that that there is a long-run and short-run relationship from FDI to Exports.

With a regression analysis using OLS, Abamu and Pietrzak (2019) found that there is a strong positive relationship between FDI and exports, an increase in FDI causes an increase in the country's exports.

Analysis results of study by Mitic and Ivić (2016) showed that a significant level of correlation between FDI and export of goods, with the stronger correlation in the case of high-tech exports.

Besides the studies supporting this view, there are still some studies that suggest that these two factors are not related to each other. With the analysis method of vector error correction model (VECM) using time series data with 40 data observations starting from the first quarter of 2010 to the fourth quarter of 2019 in Indonesia, Sumiyati (2020) found that FDI was not statistically significant in Thailand, Cambodia, Vietnam, Laos, Singapore, Indonesia, the Philippines, and Brunei. Whereas, FDI has a significant positive impact on the export of ASEAN countries. Along with the analytical method with Sumiyati (2020), with the data set collected during the period 1980-2010 in India, Sultan (2013) found a one-way causal relationship between FDI and exports in India in the long term from export to FDI. However, in the short run, they do not find any relationship between FDIs and exports.

2.2. Factors effecting exports (GDP, government expenditures, exchange rate, population, inflation, trade openness, HDI index)

In addition to being an important factor in each country's economy, GDP also has an impact on that country's export performance. There have been many studies analyzing in-depth the impact of GDP on exports from many different perspectives, and depending on the economic situation, trade policy, and economic composition of each country, these research results may be different. Numerous studies have indicated that GDP positively impacts exports in both the short and long term (BULUT & YAŞAR, 2023; Dritsakis & Stamatiou, 2018; Elshehawy et al., 2014; Long & Vuong, 2022; Sumiyati, 2020). Research conducted by Sumiyati (2020) reveals that GDP is a crucial determinant of a country's export capacity. Higher GDP means greater production capacity, which increases the economy's output and export capacity (supply side view). Furthermore, Elshehawy et al. (2014) showed that FDI increases GDP in Egypt and that exports tend to increase as income increases. Income levels are positively correlated with Egypt's export volume, i.e., as income increases, exports also increase. In addition to GDP in the exporting country, the findings also show that Egypt's exports are positively affected by GDP growth in importing countries; as income in these countries increases, Egypt's

export volume also increases. Another study in emerging economies shares the above findings, that economic growth has a positive effect on exports at lower levels of export activity, whereas the relationship between growth and exports disappears as export volume increases. This suggests that GDP can improve export performance through factors such as specialization, productivity, and production costs. However, as the volume of goods and services exported increases, these benefits may diminish over time. Therefore, in emerging market economies, shaping their strategies towards increasing the value of their products through investment in research and development (R&D), technology transfer, and specialization can increase their export potential (BULUT & YAŞAR, 2023) in international markets. However, a case study in Turkey by Yilmaz and Sensoy (2022) found that GDP has a negative impact on exports in the long run, with a 1% increase in GDP leading to a \$1.18 million decrease in exports.

Government spending is an important factor contributing to the economic development of each country. It affects many aspects of the economy, including investment in infrastructure, spending, education, etc. Research by Shvets (2024); Wilson and Benjami (2022) assumes that final goods are not subject to additional taxes when exported and are promoted if the government increases public spending in economic sectors and supports and encourages businesses to diversify exports. However, the impact of government spending on exports depends on the amount and quality of investment.

Exchange rate is an important factor affecting trade activities, especially exports. Exchange rate changes can affect the prices of goods, affecting the competitiveness of products and the profitability of enterprises. Although there are many studies on the relationship between exchange rate and exports, the results are not consistent. There are some studies that show that the real exchange rate is an important factor affecting the export performance of manufacturing and services at low and high export levels. This finding is important because it shows that the exchange rate provides a competitive advantage to emerging market economies, and they are gradually improving their production capacity (BULUT & YAŞAR, 2023; Usman, 2014). Other studies suggest that an overvalued real exchange rate is severely detrimental to export performance, while on average, a 1 percent real depreciation can increase exports by 6 to 10 percent UNCTAD (2005). Another study shows that the effective real exchange rate is an important factor behind the performance of manufacturing and services exports because it can give emerging economies a competitive advantage (AMIRKHALKHALI & DAR, 2019; BULUT & YAŞAR, 2023). Taglioni (2012) argue that exchange rate fluctuations can affect prices and cost competitiveness. Increasing challenges in domestic and foreign export markets may have the effect of reallocating resources to the most dynamic domestic producers, i.e. those who adapt more quickly to internal restructuring strategies, apply modern technology-engineering, fast production times and better product quality, improve non-price competitiveness. When production is faced with increasing returns to scale at the enterprise level, competition becomes even fiercer for domestic companies because the revaluation of the exchange rate has the effect of enhancing

additional efficiency. Because in order to improve profits, businesses will compensate for the depreciation of products due to the beneficial competition effect by increasing their output. Then the expected profit of the company will increase, leading to an increase in output, making the average cost of production decrease, improving competitive advantage in the export market.

Changes in population structure can affect the labor market and production capacity, thus shaping competitiveness in export markets. Nuroğlu (2010) argues that in the short run, population can have a positive impact on trade because it can increase human capital and increase specialization, leading to larger export volumes. However, in the long run, a higher population tends to reduce per capita income, causing consumers to spend less on goods and services, thus leading to lower production and exports. This long-term finding is similar to the study of Fohoue et al. (2024); the author also finds a negative correlation between population and exports. Population growth can reduce exports by boosting demand for domestic goods and services and reducing import volumes due to weaker economic conditions and reduced purchasing power.

Inflation is a continuous increase in the prices of goods and services over a period of time, or, in other words, the purchasing power of money decreases; consumers need to spend more money to buy the same amount of goods and services. In addition to affecting the purchasing power of consumers, inflation also affects a country's export activities. Research by BULUT and YAŞAR (2023) shows that when foreign demand is the determining factor, inflation has a positive correlation with export growth, the general price level of goods increases, and the export volume increases. However, some studies suggest that inflation has a negative impact on exports; that is, if a country's inflation increases, exports will decrease, and vice versa, when inflation decreases, exports will increase. Inflation increases the real value of money, increasing the cost of raw materials for production. As a result, export prices become more and more expensive, reducing the competitiveness of export companies in the market and making exporters less optimal in producing this item. In addition to export enterprises, the country also suffers losses due to reduced export competitiveness, leading to reduced sales and less and less foreign currency earned (Ilmas et al., 2022; Sumiyati, 2020). In addition, high inflation can be a manifestation of poor economic management (e.g., chronic government budget deficits), imperfect institutions (e.g., weak banks and financial markets), and other factors (e.g., political upheaval and social conflict) that combine to undermine export performance (Gylfason, 1999).

There are many studies on the positive relationship between trade openness and exports, but the research results are not consistent. Most of the previous studies have shown a positive relationship between trade openness and exports (AMIRKHALKHALI & DAR, 2019; Long & Vuong, 2022; Usman, 2014). Research by AMIRKHALKHALI and DAR (2019) suggests that more trade openness increases exports. In addition, a study by Usman (2014) found that trade liberalization has a positive and significant impact on sectoral export performance in Pakistan. Trade openness encourages countries to produce goods and services in which they have a comparative advantage,

creating cheaper sources of goods and increasing competitiveness in the market. This can lead to increased productivity and production volume when resources are allocated efficiently. In addition, a study in Egypt found a negative correlation between the two, with openness of importing countries not encouraging Egyptian exports but having a negative impact (Elshehawy et al., 2014).

The HDI index is not only used to measure the level of human development but also an important factor affecting the export of goods and services, contributing to the economic development of each country. Research by BAYAR and DIACONU (2022); Lukman et al. (2023) shows that training to improve skills and improve the quality of labor resources has a positive impact on exports in most countries. A workforce with experience and high professional skills will create many innovative and creative ideas for products, improve the quality of goods and services, and increase export potential with competitive prices in the market.

2.3. Limitations in previous studies

Previous studies have limited data sets with few variables and were collected over a short period of time, and consider this relationship from a simple regression perspective without considering other influencing factors so many studies do not assess the long-term impact of FDI on exports but only focus on short-term results. In addition, studies were conducted without considering the diversity of economic and political conditions among different countries, which may lead to bias in the results (Sahoo & Dash, 2022). In addition, some studies only consider this relationship from a simple regression perspective without considering other influencing factors such as macro factors and human factors (Ahmed et al., 2023).

3. Data and Methodology

3.1. Data

Data required for this study have been collected from the World Development Indicators (WDI) and Human Development Reports (HDR) over 1993–2022 of 47 Asia countries: Afghanistan, Armenia, Azerbaijan, Bangladesh, Bhutan, Brunei, Cambodia, China, Cyprus, Georgia, India, Indonesia, Iran, Iraq, Israel, Japan, Jordan, Kazakhstan, Kuwait, Kyrgyzstan, Laos, Lebanon, Malaysia, Maldives, Mongolia, Myanmar, Nepal, North Korea, Oman, Pakistan, Palestine, Philippines, Qatar, Russia, Saudi Arabia, Singapore, South Korea, Sri Lanka, Syria, Tajikistan, Thailand, Timor-Leste, Turkmenistan, United Arab Emirates, Uzbekistan, Vietnam, Yemen. According to the Organization for Economic Cooperation and Development (OECD): lower-middle income countries; middle-income countries; newly industrialized countries; OECD countries. However, due to the nature of the dataset, the countries will be divided into two groups: Upper-middle-income countries (25 countries: Armenia, Azerbaijan,

Bahrain, Brunei Darussalam, China, Cyprus, Georgia, Indonesia, Israel, Japan, Jordan, Kazakhstan, Kuwait, South Korea, Malaysia, Maldives, Mongolia, Oman, Qatar, Russian Federation, Saudi Arabia, Singapore, Thailand, Turkmenistan, United Arab Emirates), and Lower-middle-income countries (22 countries: Afghanistan, Bangladesh, Bhutan, Cambodia, India, Iraq, Iran, North Korea, Kyrgyz Republic, Lao PDR, Lebanon, Myanmar, Nepal, Pakistan, Philippines, Sri Lanka, Syrian Arab Republic, Tajikistan, Timor-Leste, Uzbekistan, Viet Nam, Yemen). The dataset consists of the following variables:

Table 1. Table of variables used in the study.

Variable type	Variable name	Definition	Source
<i>Dependent variable</i>	EXP	Exports of goods and services (current US\$)	World Development Indicators
<i>Independent variable</i>	FDI	Foreign direct investment, net inflows (BoP, current US\$)	World Development Indicators
	GDP	GDP (current US\$)	
	GOV_EX	General government final consumption expenditure (% of GDP)	
<i>Control variable</i>	EX_RATE	Official exchange rate (LCU per US\$, period average)	World Development Indicators
	POP	Population, total	
	INF	Inflation, consumer prices (annual %)	
	TRADE	Trade (% of GDP)	
	HDI	Human Development Index	Human Development Reports

3.2. Methodology

Most previous studies have used VAR models or VECM models to study the impact of FDI on exports (Ahmed et al., 2023; Bhatt, 2013; Cora & Wen, 2020; Sultan, 2013). In this study, due to the data set having many missing values, the above research models cannot be applied. Therefore, to achieve the research objective of clarifying the impact of FDI on exports in Asian countries, this article uses the research of BULUT and YAŞAR (2023) as the basis for building a research model over a period of 30 years (1993-2022). However, to overcome the limitations of previous studies, this study extends the proposed model represented by equation (1):

$$EXP_{it} = \alpha_0 + \alpha_1 FDI_{it} + \alpha_2 GDP_{it} + \alpha_3 GOV_EX_{it}$$

$$+ \alpha_4 \text{EX_RATE}_{it} + \alpha_5 \text{POP}_{it} + \alpha_6 \text{INF}_{it} + \alpha_7 \text{TRADE}_{it} + \alpha_8 \text{HDI}_{it} + \varepsilon_{it} \quad (1)$$

where:

EXP_{it} in equation (1) is the dependent variable, representing the amount of exports of goods and services and ε_{it} errors. From the independent variable, FDI_{it} represents net inflows of foreign direct investment, GDP_{it} represents GDP annual growth, GOV_EX_{it} represents annual growth of general government final consumption expenditure, EX_RATE_{it} represents official exchange rate, POP_{it} representing total population, INF_{it} represents annual growth of inflation, consumer prices, $TRADE_{it}$ represents amount of percentage of GDP of trade, and HDI_{it} represents Human Development Index. There are 47 cross sections and 30 time series in the model, the total number of observations ($47 \times 30 = 1,410$) being 1,410.

In the case of increased foreign investment, the coefficient of variable FDI_{it} (α_1) is expected to be a positive coefficient because FDI increases the country's supply capacity by increasing its capital stock. Furthermore, it also increases the competitiveness of the country's exports by increasing total factor productivity (Sultan, 2013). In the case of increased growth of GDP, the coefficient of variable GDP (α_2) is expected to be a positive coefficient because higher GDP means higher production capacity, increasing the country's export capacity (on the supply side) (Sumiyati, 2020). In the case of increased annual growth of general government final consumption expenditure, the coefficient of variable GOV_EX_{it} (α_3) is expected to be a positive coefficient because final goods may not be subject to additional taxes when exported. In addition, when the government increases spending across economic sectors, it will encourage diversification and improve export performance (Shvets, 2024; Wilson & Benjami, 2022). In the case of an increased official exchange rate, the coefficient of variable EX_RATE_{it} (α_4) is expected to be a negative coefficient because a depreciation or appreciation of the domestic currency will affect export performance. When the exchange rate increases, export volume will decrease, and when the exchange rate decreases, export volume will increase (Ilmas et al., 2022). In the case of increased total population, the coefficient of variable POP_{it} (α_5) is expected to be a positive coefficient. Since population can positively affect trade activities, it can increase human capital, increase specialization, and lead to larger export volumes (Nuroğlu, 2010). In the case of increased annual growth of inflation, the coefficient of variable INF_{it} (α_6) is expected to be a negative coefficient because inflation increases the value of money, leading to increased costs of raw materials, making goods more expensive, reducing the international competitiveness of domestic enterprises, and creating pressure for them when participating in the manufacturing sector (Ilmas et al., 2022). In the case of increased trade, the coefficient of variable $TRADE_{it}$ (α_7) is expected to be a positive coefficient because opening trade will encourage countries to produce goods and services in which they have a competitive advantage, creating a source of input goods. cheaper, expanding competition opportunities in the international market (Usman, 2014). In the case of increased HDI, the coefficient of variable HDI_{it} (α_8) is expected to be a positive coefficient because a more educated and skilled workforce will produce better

quality goods and services, enhancing competitive advantage and providing more export opportunities (Lukman et al., 2023).

Multicollinearity (VIF test): Multicollinearity is the presence of two or more independent variables in a regression model with a high degree of correlation. If there is no linear relationship between predictor variables, they are said to be orthogonal (Daoud, 2009). When one or more independent variables in a regression model are strongly correlated with each other, it can make the regression results misleading and unreliable. Not only that, when the model has multicollinearity, it will make the regression coefficients difficult to interpret and increase the variability of the estimates, creating many obstacles in determining the impact of each independent variable. set up the dependent variable.

The VIF is a tool to measure and quantify how much the variance is inflated. VIFs are usually calculated by the software as part of regression analysis and will appear in the VIF column as part of the output. To interpret the value of VIF, the following rule is used in the table below:

Table 2. VIF interpretation

VIF -value	Conclusion
$VIF < 1$	Not correlated
$1 < VIF < 5$	Moderately correlated
$VIF > 5$	Highly correlated

Source : Daoud (2009).

F-test: For panel data, the fixed effects model should be used when the effects are cross-sectionally or time-specific. If these effects are not significant, the Pooled OLS model is appropriate. Therefore, it is necessary to evaluate whether a Fixed Effects Model is necessary or a pooled OLS model is appropriate. This can be assessed using an F-test, which determines whether the inclusion of fixed effects improves the fit of the model and compares the results of the Fixed Effects model with the results of pooled OLS to determine which model is appropriate (Rehal, 2024). The null hypothesis and alternative hypothesis of the F test are stated as follows:

H_0 : Fixed effects are jointly zero (use Pooled OLS)

H_A : Fixed effects are significant (use Fixed Effects Model)

According to the null hypothesis, all fixed effects are jointly equal to zero or not statistically significant. Thus, if we fail to reject the null hypothesis, the Pooled OLS model is appropriate for use. On the other hand, if we do reject the null hypothesis, it indicates that we should utilize the Fixed Effects Model, as this suggests that the cross-sectional or time-specific effects are statistically significant (Rehal, 2024).

Hausman test: The Hausman test is a statistical method used to compare two different regression models in panel data analysis, specifically between the Fixed Effects Model (FEM) and the Random Effects Model (REM). This method helps to determine which

model is more suitable for the available data (H.Baltagi, 2012; Sutikno et al., 2017). We can further state the null and alternate hypotheses of the Hausman test as follows:

H_0 : correlation $(\mu_i, X_{it}) = 0$ (REM model)

H_1 : correlation $(\mu_i, C_{it}) \neq 0$ (FEM model)

If $p\text{-value} \leq 0.05$, then hypothesis H_0 is rejected, meaning that FEM is a better model than REM. But if $p\text{-value} > 0.05$, then hypothesis H_0 is accepted. This points out that the best model between FEM and REM is REM (H.Baltagi, 2012; Sutikno et al., 2017).

Heteroscedasticity (Breusch-Pagan test): In statistics, heteroskedasticity (or heteroscedasticity) happens when the standard deviations of a predicted variable, monitored over different values of an independent variable or as related to prior time periods, are non-constant. This violates one of the fundamental assumptions of linear regression: that the errors must be homoscedastic. When heteroscedasticity exists in the model, the loss in efficiency when using ordinary least squares (OLS) can be large, more importantly, the bias in the estimated standard errors can make the results unreliable, leading to bias in statistical hypothesis testing and invalid inferences. In statistics, the Breusch-Pagan test, developed by Trevor Breusch and Adrian Pagan in 1979, is used to test for heteroscedasticity in linear regression models. This method is based on the Aitchison and Silvey Lagrangian kernel (LM) test, also known as Rao's efficient score test, which tests whether the variance of the errors from the regression depends on the values of the independent variables. In that case, heteroscedasticity is present (Breusch & Pagan, 1979). We can further state the null and alternative hypotheses of the Breusch-Pagan test as follows:

H_0 : The variance is constant.

H_1 : The variance is not constant.

If the test statistic has a $p\text{-value}$ below an appropriate threshold (e.g, $p < 0.05$), then the null hypothesis of homoskedasticity is rejected and heteroskedasticity is assumed.

Serial correlation (Wooldridge test): Serial correlation is a common problem in regression model analysis, especially in time series data and panel data. This phenomenon occurs when the errors in the regression model are not independent of each other; that is, the error value at the current time point has a relationship with the error value at the previous time point. This can cause bias in the estimation of regression parameters, making statistical tests inaccurate (Drukker, 2003). The Wooldridge test is used to detect serial correlation in panel data. This test helps to check whether the errors are independent or not by checking the regression series between the errors. Specifically, the Wooldridge test can be presented through the following hypotheses:

H_0 : The errors are independent (no serial correlation).

H_a : Non-independent errors (serial correlation).

If the test statistic has a $p\text{-value}$ lower than the appropriate threshold (e.g, $p < 0.05$), reject the null hypothesis, which means that there is a serial correlation phenomenon. If $p > 0.05$, accept the null hypothesis.

SGMM method: Generalized Method of Moments (GMM) to estimate and address issues related to heteroscedasticity and serial correlation within the model, owing to the characteristics of the panel data utilized in this research. Arellano and Bond (1991) introduced a GMM technique for estimating regression when there is the presence of lagged dependent variables. However, Blundell et al. (2000) demonstrated that Difference GMM (D-GMM) may lead to bias in small samples, particularly when the series exhibit high persistence. In such cases, System GMM (S-GMM) is regarded as the preferable option, Arellano and Bover (1995). The system generalized method of moments (SGMM) is an estimation method commonly used in panel data analysis, especially in dynamic regression models, where there is a bivariate relationship between the independent and dependent variables. Unlike the traditional dynamic GMM method, SGMM is a combination of static and dynamic regression, which enables research to detect hidden relationships in the data while ensuring that the estimation results are consistent and efficient.

One of the greatest advantages of SGMM is its ability to handle endogeneity and heteroscedasticity issues, which are common problems in research. By using model terms and moments from both past and current variables, SGMM can provide more accurate estimates of model parameters than other estimation methods. According to Blundell and Bond (1998), SGMM can improve the accuracy and reliability of estimates when used in complex panel data models, especially when there is heteroscedasticity in the error term.

The SGMM model can be written as follows:

$$Y_{it} = \alpha + \beta Y_{it-1} + \gamma X_{it} + \epsilon_{it}$$

Where Y_{it-1} is the lag variable of the dependent variable, X_{it} is the independent variable, and ϵ_{it} is the error term.

Instruments from the lagged variables of both y and x are used to create moment conditions, allowing control for endogeneity and heteroskedasticity issues. An advantage of SGMM is its ability to improve the robustness of the estimates through the use of a system of equations, thereby reducing bias in parameter estimates. Research has shown that SGMM can provide more accurate estimates in dynamic regression models than the simple GMM method, especially when heteroskedasticity in the errors is a problem (Blundell & Bond, 1998).

4. Result

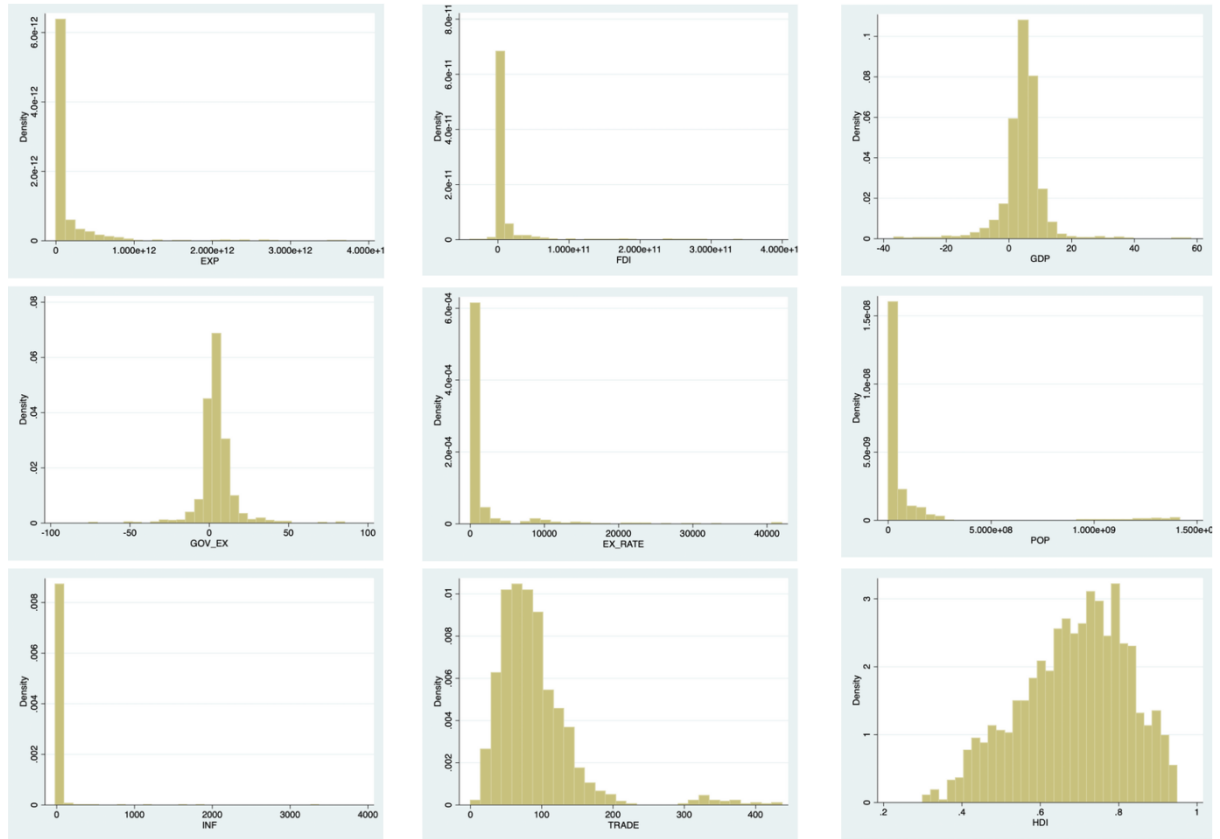
4.1. Variable definitions and summary statistics

Table 3. Variable definitions and summary statistics

Variable	Description	Obs	Mean	Std. Dev.	Min	Max
EXP	Exports of goods and services (current US\$)	1,248	1.22e+11	3.13e+11	693281	3.72e+12
FDI	Foreign direct investment, net inflows (BoP, current US\$)	1,371	7.78e+09	2.76e+10	-3.98e+10	3.44e+11
GDP	GDP growth (annual %)	1,367	4.507488	6.495791	-36.65815	58.07811
GOV_EX	General government final consumption expenditure (annual % growth)	974	4.508044	10.62097	-75.98515	85.35944
EX_RATE	Official exchange rate (LCU per US\$, period average)	1,331	1456.992	4509.878	.0103574	42000
POP	Population, total	1,410	8.77e+07	2.53e+08	245845	1.42e+09
INF	Inflation, consumer prices (annual %)	1,217	16.37934	129.7099	-16.11733	3373.759
TRADE	Trade (% of GDP)	1,247	89.30817	55.98112	.0209992	437.3267
HDI	Human Development Index	1,307	.6867154	.1350555	.3	.949

Source: Author's calculation

Figure 1. Histogram of varibales used in model



Sample size

The dataset includes 47 countries from the Asia region collected over 30 years (1993-2022). Due to data gaps during the collection process, the variables have differing numbers of observations. The highest number of observations is 1,410, while the lowest is 974. Although this sample size is still relatively small, it is sufficient to evaluate the research impact objective.

Dependent variable

According to equation (1) above, the variable EXP is the dependent variable. Exports of goods and services represent the value of all goods and other market services provided to the rest of the world. They include the value of merchandise, freight, insurance, transport, travel, royalties, license fees, and other services, such as communication, construction, financial, information, business, personal, and government services. They exclude compensation of employees and investment income (formerly called factor services) and transfer payments. From the results of Table 3, variable EXP has a total of 1,248 observations and 162 are missing. Thus, the smallest, largest and mean values are 1.22e+11(US\$), 693281(US\$), and 3.72e+12 (US\$), respectively, with a standard deviation of 3.13e+11 (US\$). This shows that there is a huge difference between the smallest value and the largest value of exports of goods and services in countries in the Asian region in 30 years. There are some countries with limited export economies, but there

are also several countries with strongly developed export economies. Besides, the standard deviation of export value is also high, showing that there are clear differences between countries in the region.

Independent variables

According to equation (1) above, the variable FDI is the independent variable. Foreign direct investment refers to direct investment equity flows in the reporting economy. It is the sum of equity capital, reinvestment of earnings, and other capital. Direct investment is a category of cross-border investment associated with a resident in one economy having control or a significant degree of influence on the management of an enterprise that is resident in another economy. Ownership of 10 percent or more of the ordinary shares of voting stock is the criterion for determining the existence of a direct investment relationship. From the results of Table 3, variable FDI has a total of 1,371 observations and 39 missing. Thus, the smallest, largest and mean values are $-3.98e+10$ (US\$), $3.44e+11$ (US\$), and $7.78e+09$ (US\$), respectively, with a standard deviation of $2.76e+10$ (US\$). Thereby, there are still years when FDI has a negative value because the investment capital flow into the country is smaller than the investment capital flow abroad. This can happen when there is global geopolitical instability or unfavorable trade policies. Besides, the standard deviation of foreign investment capital is also high, showing that there are clear differences between countries in the Asian region over the past 30 years.

Control variables

According to equation (8) above, we have 7 independent variables as follows:

The first is the variable economic growth (GDP). Annual percentage growth rate of GDP at market prices based on constant local currency. Aggregates are based on constant 2015 prices, expressed in U.S. dollars. From the results of Table 3, variable GDP has a total of 1,368 observations and 42 missing. Thus, the smallest, largest and mean values are 4.507488, -36.65815, and 58.07811, respectively, with a standard deviation of 6.495791. Thereby, the average economic growth rate of Asian countries over the past 30 years is 4.51%. Additionally, the standard deviation is quite high (6.5%) with the lowest value being -36.66% and the highest value being 58.08%, which shows that there are differences between countries. There are countries with negative GDP growth that may be affected by an economic crisis, severe due to economic crisis, unfavorable financial conditions, natural disasters, etc.

The second is the variable government expenditure (GOV_EX). Annual percentage growth of general government final consumption expenditure based on constant local currency. Aggregates are based on constant 2015 prices, expressed in U.S. dollars. General government final consumption expenditure (general government consumption) includes all government current expenditures for purchases of goods and services

(including compensation of employees). It also includes most expenditures on national defense and security but excludes government military expenditures that are part of government capital formation. From the results of Table 3, variable GOV_EX has a total of 974 observations and 436 missing values. Thus, the smallest, largest and mean values are 4.5080447, -75.98515, and 85.35944, respectively, with a standard deviation of 10.62097. Thereby, the average government spending of Asian countries in the past 30 years is 4.51% of GDP. Besides, the standard deviation is quite high (10.62%) with the lowest value of government spending being -75.99% of GDP and the highest value of government spending being 85.36%. This shows that there are differences between countries in the region due to different economic and political contexts as well as different development strategies. Therefore, there are countries where the government increases public investment to recover the economy after economic crises, natural disasters, epidemics, etc., and there are also countries that implement austerity trends to stabilize the economy. macroeconomics such as inflation, etc.

The third is the variable official exchange rate (EX_RATE) refers to the exchange rate determined by national authorities or to the rate determined in the legally sanctioned exchange market. It is calculated as an annual average based on monthly averages (local currency units relative to the U.S. dollar). From the results of Table 3, variable EX_RATE has a total of 1,331 observations and 79 missing values. Thus, the smallest, largest and mean values are 0.0103574, 42000, and 1456.992, respectively, with a standard deviation of 4509.878. This shows that there is a huge difference between the smallest and largest values of official exchange rates in Asian countries over the past 30 years. Besides, the standard deviation of the official exchange rate is also high, showing that there are clear differences between countries in the region due to different monetary policies and economic conditions.

The fourth is the variable population (POP) calculated by the total population of each country. From the results of Table 3, variable POP has a total of 1,410 observations. Thus, the smallest, largest and mean values are $8.77e+07$ (people), 245845 (people), and $1.42e+09$ (people), respectively, with a standard deviation of $2.53e+08$ (people). This shows that there is a huge difference between the smallest and largest values of population in Asian countries over the past 30 years. Besides, the standard deviation of the total population is also high, showing the diversity in population size among countries in the region.

The fifth is the variable inflation (INF) as measured by the consumer price index, which reflects the annual percentage change in the cost to the average consumer of acquiring a basket of goods and services that may be fixed or changed at specified intervals, such as yearly. From the results of Table 3, variable INF has a total of 1,217 observations and 193 missing values. Thus, the smallest, largest and mean values are -16.11733 (%), 3373.759 (%), and 16.37934 (%), respectively, with a standard deviation of 129.7099 (%). Thereby, the lowest inflation level is -16.12%, which means that some countries are facing deflation. Deflation can lead to an economic recession, causing the amount of produced and consumed goods to decline, affecting export performance.

The sixth is the variable trade openness (TRADE) calculated by the sum of exports and imports of goods and services measured as a share of gross domestic product (% of GDP). From the results of Table 3, variable TRADE has a total of 1,247 observations and 163 missing values. Thus, the smallest, largest and mean values are 0.0209992 (%), 437.3267 (%), and 89.30817 (%), respectively, with a standard deviation of 55.98112 (%). Additionally, the standard deviation is quite high (55.98112 %) with the lowest value being 0.0209992% and the highest value being 437.3267%, which shows that there are differences between countries. There are some countries whose economic growth depends heavily on international trade, but there are some countries that depend less on it.

The seventh is the variable Human Development Index (HDI), calculated based on gross national income divided by population, life expectancy at birth, and average years of schooling. It provides a snapshot of human development. From the results of Table 3, variable HDI has a total of 1,307 observations and 103 missing values. Thus, the smallest, largest and mean values are 0.3, 0.949, and 0.6867154, respectively, with a standard deviation of 0.1350555. With a standard deviation of 0.1351, it shows that there is a difference in the human development index in Asian countries over the past 30 years. With the smallest value of 0.3 (low HDI level < 0.550) and the highest value of 0.949 (very high HDI level > 0.8). A low HDI may be because the country is still poor and has not really invested much in education, health, infrastructure, etc. On the contrary, the HDI is high.

From the statistical table describing the variables combined with the histogram, only three variables: GDP, GOV_EX and HDI have a normal distribution; the remaining variables, such as: EXP, FDI, EX_RATE, POP, INF and TRADE all have a non-normal distribution. Therefore, replacing the non-normally distributed variables by taking the logarithm of each variable, the new research model is:

$$\ln_EXP_{it} = \alpha_0 + \alpha_1 \ln_FDI_{it} + \alpha_2 \ln_GDP_{it} + \alpha_3 GOV_EX_{it} + \alpha_4 \ln_EX_RATE_{it} + \alpha_5 \ln_POP_{it} + \alpha_6 \ln_INF_{it} + \alpha_7 TRADE_{it} + \alpha_8 HDI_{it} + \varepsilon_{it} \quad (2)$$

4.2. Inspection results

4.2.1. Results of correlation

After collecting the complete data set, we tested the correlation between variables in the model and obtained the following table:

Table 4. Correlation between variables in the model

	ln_EXP	ln_FDI	GDP1	GOV_EX1	ln_EX_RATE
ln_EXP	1.0000				
ln_FDI	0.8182*	1.0000			
GDP1	-0.0081*	0.0778*	1.0000		

GOV_EX1	-0.0228*	-0.0122*	0.2864*	1.0000	
ln_EX_RATE	-0.0063*	0.0385*	0.0474*	0.0556*	1.0000
ln_POP1	0.5645*	0.4562*	0.0118*	-0.0199*	0.3991*
ln_INF	-0.2724*	-0.2157*	-0.0445*	-0.1030*	0.2739*
ln_TRADE	0.0846*	0.0219*	0.0331*	0.0771*	-0.2814*
HDI	0.5424*	0.4913*	-0.0925*	-0.0419*	-0.3164*

	ln_POP1	ln_INF	ln-TRADE	HDI
ln_POP1	1.0000			
ln_INF	0.1577*	1.0000		
ln_TRADE	-0.4584*	-0.2476*	1.0000	
HDI	-0.2674*	-0.4062*	0.2933*	1.0000

Source: Author's calculation

Based on the results in Table 4, we see that the correlation coefficients between the variables are all less than 80%; only the correlation coefficient between the variables ln_FDI and ln_EXP is high (over 80%) and still statistically significant. Because the results are only predictive, the FDI variable is still retained to estimate the research model. If the model has autocorrelation, it will be resolved in the next section.

4.2.2. Results of Multicollinearity test

Table 5. Results of Multicollinearity test

	VIF	1/VIF
ln_POP	3.78	0.264516
ln_FDI	3.06	0.326665
HDI	2.64	0.378163
ln_TRADE	2.07	0.483669
ln_EX_RATE	1.34	0.746517
ln_INF	1.28	0.779846
GDP	1.21	0.823339
GOV_EX	1.14	0.876317
Mean VIF	2.07	

Source: Author's calculation

Based on the results of multicollinearity testing in the model in the table, the VIF of all variables is less than 5. Therefore, the predicted multicollinearity phenomenon in the model is quite low.

4.2.3. Result of OLS, FEM, REM model

We use the data set for OLS, Random Effect Model (REM), Fixed Effect Model (FEM). To choose the most suitable model, we consider the results of the F-test and Hausman test. In the table, after performing the F-test with $\text{Prob} > F = 0.0000$, this proves that the FEM model is more suitable than the OLS model. We then use the Hausman test to choose between the FEM and REM models. With the result of the Hausman test being $\text{Prob} > \chi^2 = 0.0000$, we accept the hypothesis H_0 , meaning that the FEM model is more suitable for the data set that we are using than the REM model. The above results are shown in Table 5 below.

Table 6. OLS, REM, FEM regression results

	Reg	Rem	Fem
ln_FDI	0.214*** (10.44)	0.0657*** (5.98)	0.0682*** (5.89)
ln_GDP	-0.0189*** (-2.88)	0.00780*** (2.80)	0.00914*** (3.11)
ln_GOV_EX	0.00323 (1.18)	0.000244 (0.22)	0.000567 (0.49)
ln_EX_RATE	-0.0391*** (-4.32)	-0.198*** (-7.18)	-0.130*** (-5.50)
POP	0.872*** (33.72)	-0.169 (-1.31)	0.787*** (12.38)
ln_INF	-0.114*** (-4.58)	-0.00150 (-0.12)	-0.00435 (-0.33)
TRADE	0.720*** (11.64)	0.501*** (9.04)	0.562*** (9.77)
HDI	8.848*** (27.71)	13.75*** (32.46)	11.22*** (33.86)
Constant	-4.163*** (-8.42)	14.34*** (7.05)	-0.582 (-0.56)
N	766	766	766
R-squared	0.890	0.857	
F-test	Prob > F = 0.0000		
Hausman test	Prob > χ^2 = 0.0000		
Breusch – Pagan test	Prob > χ^2 = 0.0000		
Wooldridge test	Prob > F = 0.0000		

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's calculation

With the results of FEM, the regression equation (2) has 6 independent variables that have a significant impact on exports at the 10% level. After performing the Wooldridge test and the Breusch-Pagan test, the results of the FEM model are no longer accurate because the data set has heterogeneity of variance and autocorrelation. Besides, there are some studies that suggest that there is endogeneity between FDI and exports (Ahmed et al., 2023; Bhatt, 2013; Bouras & Raggad, 2015; Liu & Graham, 1998; Sahoo & Sethi, 2020; Sultan, 2013). Therefore, to overcome these limitations, we perform regression of the dataset based on another regression method and will be presented in the next section.

4.3. Results

If heteroskedasticity and autocorrelation exist in the model, the FGLS model will be performed (Pham et al., 2022). Although the FGLS model can deal with heteroskedasticity problems and autocorrelation, bias relating to endogeneity still exists. Then, the SGMM method will be applied. The SGMM method proposed by Arellano and Bond (1991) is suitable because it solves the defects of the panel data model such as autocorrelation, heteroscedasticity, and endogeneity (Pham et al., 2022). The GMM model, which is generally used for panel data, provides consistent results in the presence of different sources of endogeneity, namely “unobserved heterogeneity, simultaneity and dynamic endogeneity. The GMM model removes endogeneity by “internally transforming the data” – transformation refers to a statistical process where a variable’s past value is subtracted from its present value (UllahSubhan et al., 2018). Therefore, to overcome the problems of heteroscedasticity, autocorrelation and endogeneity in the FEM model in the above section, this study will use the S-GMM model to estimate the impact of foreign direct investment (FDI) on exports. The use of the System GMM method ensures compliance with the constraints; the validity of the instrumental variables is assessed through the Arellano-Bond, Hansen, and Sargan tests. The Arellano-Bond test (AR1) is used to detect first-order serial correlation, and (AR2) is used to detect second-order serial correlation in the model. Therefore, the results of the AR(1) test will not be considered when the second-order serial correlation test (AR2) is used. The Hansen test evaluates the null hypothesis (H_0) that the instrumental variables are strongly exogenous, indicating that there is no correlation with the errors. Therefore, the larger the *p-value* of the Hansen test, the more suitable the instrumental variables used are for the research model (Hansen, 1982). The research results are presented in Table 6, including the overall sample (47 countries in the Asian region) and two separate groups: upper middle-income countries, and lower middle-income countries.

Table 7. SGMM results

	47 countries in Asia	UMIs	LMIs
L.ln_EXP	0.617 ***	0.687***	0.844***

	(6.92)	(6.97)	(5.08)
ln_FDI	0.0625**	0.0789**	0.105**
	(2.02)	(2.21)	(2.09)
GDP	0.0167***	0.0174**	0.0239***
	(2.60)	(1.97)	(4.55)
GOV_EX	-0.00968	-0.0125	-0.00243
	(-1.29)	(-0.91)	(-0.36)
ln_EX_RATE	-0.0199*	-0.0180	0.0191
	(-1.67)	(-1.36)	(0.91)
ln_POP	0.342***	0.245**	0.0660
	(3.95)	(2.16)	(0.42)
ln_INF	-0.0381*	-0.0136	0.0335
	(-1.67)	(-1.33)	(0.77)
ln_TRADE	0.331***	0.277**	0.0317
	(4.55)	(2.54)	(0.32)
HDI	3.397***	3.101*	0.102
	(3.62)	(1.81)	(0.10)
Constant	-1.554**	-1.566*	-0.0850
	(-2.33)	(-1.73)	(-0.19)
Number of obs	685	366	319
Number of groups	39	20	19
Number of instruments	14	16	12
AR (1)			
(p-value)	0.008	0.006	0.005
AR (2)			
(p-value)	0.077	0.190	0.287
Sargan test			
(Prob > chi2)	0.145	0.967	0.777
Hansen test			
(Prob > chi2)	0.355	0.894	0.844

t statistics in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Source: Author's calculation

The results pass necessary diagnostic tests like serial correlation order one and two (AR (1) and AR (2)) and the exogeneity test of instruments (Hansen test). The study uses two-step S-GMM regression with lagged values of both dependent (starting from lag 3 for the dependent variable) and independent variables (lag 3 for GOV_EX variable) are used as instruments. Other than FDI and exports, other independent variables are treated as exogenous. The model uses an additional robust option to

calculate the standard errors of the model estimates under the assumption that there is heteroskedasticity or that there may be autocorrelation in the errors. This helps ensure that the estimates are not biased due to unsatisfied assumptions about the distribution of the errors, and the optimal number of instruments is derived using the collapse option.

From the results of the table above, the lagged export coefficient (L.ln_EXP) in all 3 samples has a positive coefficient with a high significance level at 1%, the coefficient of lagged export value is less than one. This shows the stable development of exports over the past 30 years. For the general research sample of 47 countries in the Asian region, exports in the previous year have a positive significance for exports in the following year with a positive coefficient of 0.617 with statistical significance at the level of 1%. This means that under the condition that other factors remain unchanged, when exports in the previous year increase by 1 unit, exports in the following year in Asian countries increase by 0.617 units, or, in other words, when exports in the previous year increase by 1%, exports in the following year in Asian countries increase by 0.617%. Next, for the individual research samples, the export volume in the previous year has a positive significance for the export volume in the following year and has different impact scales among the 2 groups of countries, upper-middle income countries (UMIs) (coefficient: 0.687), and lower-middle income countries (LMIs) (coefficient: 0.844). Thus, under the condition that other factors remain constant, when the export volume in the previous year increases by 1%, the export volume in the following year in upper-middle income, and lower-middle income countries increases by 0.687 percent, and 0.844 percent.

For the foreign direct investment (FDI) variable, it has a positive and statistically significant coefficient (below 5%) for the entire sample and for 2 individual samples: UMIs (Upper-Middle Income countries), LMIs (Lower-Middle Income countries). This result is consistent with the initial expectation set for the study and is similar to the results of some previous studies by Cora and Wen (2020); Liam T. Kane and Alexander R. Vaccaro (2020); Liu and Graham (1998); UNCTAD (2005). Because FDI not only increases export volume but also increases the value of goods. Because most multinational corporations have a lot of production experience, own many advanced production technologies, and are more closely linked to the supply chain of the parent company than domestic companies. At the same time, the spillover effects generated by the production activities of multinational corporations can positively affect the productivity of domestic enterprises by improving their operating and production processes. This in turn allows consumers to buy higher-quality goods and services, increasing the export turnover of the FDI-receiving country (BAYAR & DIACONU, 2022). For the general research sample of 47 countries in the Asian region, foreign direct investment has a positive significance for the export of goods and services with a positive coefficient of 0.0625 with statistical significance at the 5% level. This means that, with other factors remaining constant, when foreign investment increases by 1%, the export of goods and services in Asian countries increases by 0.0625%. Next, for the individual research samples, foreign investment capital is positive and has different

impact scales among the 2 groups of countries: upper-middle income countries (UMIs) (coefficient: 0.0789), and lower-middle income countries (LMIs) (coefficient: 0.105). Thus, under the condition that other factors remain unchanged, when foreign investment capital increases by 1%, exports of goods and services increase in upper-middle income countries, and lower-middle income countries by 0.0789 percent, and 0.105 percent.

For the economic growth rate variable (GDP), it has a positive and statistically significant coefficient for the entire sample and for 2 individual samples: UMIs (Upper-Middle Income countries), LMIs (Lower-Middle Income countries). This is also consistent with the results of some previous studies of BULUT and YAŞAR (2023); Elshehawy et al. (2014); Long and Vuong (2022); Yilmaz and Sensoy (2022). Because in emerging market economies, businesses that invest heavily in R&D, technology, and more specialized and modern production processes will increase productivity as well as product quality, thereby increasing competitive advantage in the international market and increasing export potential (BULUT & YAŞAR, 2023). For the general research sample of 47 countries in the Asian region, economic growth rates has a positive significance for the export of goods and services with a positive coefficient of 0.0167 with statistical significance at the 1% level. This means that, with other factors remaining constant, when economic growth rates increases by 1%, the export of goods and services in Asian countries increases by 0.0167%. Next, for the individual research samples, economic growth rate is positive and has different impact scales among the 2 groups of countries: upper-middle income countries (UMIs) (coefficient: 0.0174 with statistical significance at the 5% level), and lower-middle income countries (LMIs) (coefficient: 0.0239 with statistical significance at the 1% level). Thus, holding other factors constant, when economic growth rates increase by 1%, exports of goods and services increase in upper-middle income countries, and lower-middle income countries by 0.0174 percent, and 0.0239 percent.

For the government final expenditure variable (GOV_EX), it has a negative coefficient contrary to initial expectations and is not statistically significant in all three samples: 47 Asia countries, UMIs, and LMIs.

For the exchange rate variable (EX_RATE), it is not statistically significant for upper-middle income countries and lower-middle income countries. However, this variable has a negative coefficient (coefficient: -0.0199) with statistical significance at the 10% level for the overall sample of 47 countries. This is consistent with the initial expectation and also consistent with the results of some previous studies by Ilmas et al. (2022); UNCTAD (2005). When the exchange rate is overvalued, or the domestic currency appreciates, it will cause the price of exported goods to increase in the international market, thereby causing consumers to switch to buying cheaper goods, leading to businesses having to reduce production, causing disadvantages for export performance. Therefore, when the exchange rate increases, exports will decrease, and when the exchange rate decreases, exports will increase (Ilmas et al., 2022; UNCTAD, 2005). Thus, assuming other factors remain constant, when government spending increases by 1%, exports in 47 countries decrease by 0.0199 (%).

For, the total population variable (POP) variable, it is not statistically significant for lower-middle income countries. However, it has a positive and statistically significant coefficient for the entire sample and UMIs (Upper-Middle Income countries). In the short term, the total population has a positive effect on export performance because population growth increases the working-age population, increases the level of specialization, and the quantity of exported goods increases. However, if we consider the long term, a high population will reduce GDP per capita, leading to lower income and reduced consumer spending, leading to manufacturers producing less to reduce inventories and exports also decreasing (Nuroğlu, 2010). For the overall sample of 47 countries in the Asian region, total population has a positive effect on exports with a positive coefficient of 0.342, which is statistically significant at the 1% level. This means that, holding other factors constant, when total population increases by 1%, exports in Asian countries increase by 0.342%. Next, for upper-middle income countries, total population has a positive significance and statistically significant at 5% level (coefficient: 0.245). Thus, under the condition that other factors remain constant, when the total population increases by 1 (%), exports in upper-middle income countries increase by 0.245 percent.

For the inflation variable (INF), it is not statistically significant for upper-middle income countries and lower-middle income countries. However, this variable has a negative coefficient (coefficient: -0.0381) with statistical significance at the 10% level for the overall sample of 47 countries. Because when inflation increases, operating and production costs also increase, thus making exporters less optimal in producing this item. When commodity prices become more expensive and affect the slow purchasing power of people, it will reduce the competitiveness of domestic companies' export goods and create barriers for them to enter the manufacturing sector. In addition to the negative impacts on business production activities, high inflation can be a sign of ineffective economic management or due to factors such as economic recession, political fluctuations, ... thereby reducing export performance (Gylfason, 1999; Ilmas et al., 2022; Sumiyati, 2020). Thus, assuming other factors remain constant, when inflation increases by 1%, exports in 47 countries decrease by 0.0381 (%).

For the trade openness variable (TRADE), although it has a positive coefficient, it is not statistically significant for lower-middle income countries (LMIs). However, TRADE has a positive coefficient and is quite statistically significant (1%) for the general sample of 47 countries and upper-middle income countries. Because trade liberalization creates opportunities for countries to cooperate, support, and encourage the production of goods in which countries have a comparative advantage, creating a path for exports with lower input prices, improving competitiveness in the global market (Usman, 2014). For the overall sample of 47 countries in the Asian region, trade openness has a positive coefficient of 0.331 on exports, which is significant at the 1% level. This means that, holding other factors constant, a 1% increase in trade openness increases exports in Asian countries by 0.331%. Next, for upper-middle income countries, trade openness has a positive significance and statistically significant at 5%

level (coefficient: 0.277). Thus, under the condition that other factors remain constant, when trade openness increases by 1 (%), exports in upper-middle income countries increase by 0.277 percent.

For the human development index (HDI) variable, although it has a positive coefficient, it is not statistically significant for lower-middle income countries (LMIs). However, HDI has a positive coefficient with statistically significant for the general sample of 47 Asia countries and upper-middle income countries (UMIs). This result is consistent with the initial expectation of the study because when the workforce participates in training programs, it can help them improve their working skills and expertise and continuously develop new products, services, and production methods to meet market needs. Thereby gradually improving the quality of products and services, increasing competitive advantages in the export market (BAYAR & DIACONU, 2022; Lukman et al., 2023). For the general research sample of 47 countries in the Asian region, HDI has a positive significance for exports with a positive coefficient of 3.397 with statistical significance at the 1% level. This means that, with other factors remaining constant, when HDI increases by 1 unit, exports in Asian countries increase by 3.397 (%). Next, for the upper-middle income countries (UMIs) has coefficient at 3.101 with statistical significance at the 10% level. Thus, under the condition that other factors remain constant, when HDI increases by 1 unit, exports in upper-middle income countries increase by 3.101%.

5. Conclusion and policy implication

5.1. Conclusion

The study results found a positive correlation between foreign direct investment (FDI) and exports in Asian countries in general and upper-middle income countries, lower-middle income countries, OECD countries, and emerging countries in particular. Along with improving production capacity, technology, and product quality, FDI also plays a pivotal role in augmenting enterprise competitiveness, opening up many potential opportunities in the export market. However, the study also shows that depending on the socio-economic development context, factors such as economic development policies, investment environment, etc., the impact of FDI on exports may not be uniform across groups of countries. Notably, FDI exerts the most significant impact in upper-middle income countries, whereas its effect is comparatively diminished in emerging economies. In addition to FDI, factors such as economic growth rate, government spending, exchange rate, total population, inflation, trade openness, and FDI also have an impact on exports. Among them, factors such as economic growth, total population, inflation, trade openness, and HDI index all have positive impacts on exports, while exchange rate and government spending have negative impacts on exports in some research samples.

5.2. Policy implications

First, to promote foreign direct investment (FDI) in the country, it is necessary to build a potential investment environment for enterprises. Each country should implement tax incentives for FDI enterprises, such as reducing corporate income tax in the first years of operation...or exempting or reducing fees related to administrative procedures related to FDI.

In addition to policies to increase foreign direct investment, policies to stimulate GDP growth, improve the Human Development Index (HDI) and promote trade also contribute to improving export performance. To stimulate economic growth, each country should continue to increase investment in developing transport infrastructure systems and investing in, building digital infrastructure and fundamental technologies for transformation to support the export of goods. In addition, with the problem of global energy shortage, investing in sustainable energy sources to ensure the stability of energy supply for manufacturing enterprises also plays an equally important role. Along with socio-economic development policies, each country needs to invest in education, training, and health care programs to improve the HDI index and develop modern technological and technical skills for the workforce. In terms of trade openness, each country needs to actively participate in Free Trade Agreements and new generation Free Trade Agreements to create conditions for domestic enterprises to have the opportunity to participate in potential markets around the world, as well as help consumers access a rich source of quality goods at reasonable prices. In addition, it is necessary to create more trade promotion activities and trade fairs to help enterprises find stable sources of raw materials for production, diversify export markets, and, at the same time, learn, approach, and connect with new markets and partners, take advantage of the benefits of Free Trade Agreements (FTAs), strengthen international market connections, and participate deeply in global value chains.

In addition to export promotion policies, each country needs to have strategies to manage and prevent risks to stabilize the economy. Regarding inflation, it is necessary to improve the effectiveness of coordinating fiscal and monetary policies in managing money supply, interest rates, and balancing the amount of money in and out to maintain price stability. And finally, in parallel with controlling inflation, each country needs to monitor and manage exchange rates to minimize negative impacts on exports, create a flexible foreign exchange trading environment, and help businesses adapt to currency fluctuations.

Appendix A

List of countries

1. Upper-Middle Income Countries (UMIs):

Armenia, Azerbaijan, Bahrain, Brunei Darussalam, China, Cyprus, Georgia, Indonesia, Israel, Japan, Jordan, Kazakhstan, Kuwait, South Korea, Malaysia, Maldives, Mongolia, Oman, Qatar, Russian Federation, Saudi Arabia, Singapore, Thailand, Turkmenistan, United Arab Emirates.

2. Lower Middle-income Countries (LMIs): Afghanistan, Bangladesh, Bhutan, Cambodia, India, Iraq, Iran, North Korea, Kyrgyz Republic, Lao PDR, Lebanon, Myanmar, Nepal, Pakistan, Philippines, Sri Lanka, Syrian Arab Republic, Tajikistan, Timor-Leste, Uzbekistan, Viet Nam, Yemen.

Appendix B

s

Table 1. Pooled OLS results

	UMIs	LMIs
ln_FDI	0.0592** (2.37)	0.215*** (8.47)
GDP	-0.0139** (-2.16)	-0.000863 (-0.09)
GOV_EX	0.00974** (2.28)	-0.000720 (-0.26)
ln_EX_RATE	- 0.0813*** (-6.99)	0.108*** (8.55)
ln_POP	0.992*** (36.45)	0.849*** (23.21)
ln_INF	- 0.0889*** (-3.54)	0.0452 (1.17)
ln_TRADE	11.71*** (25.09)	5.073*** (12.84)
HDI	0.871*** (13.52)	0.418*** (4.57)
Constant	-5.562*** (-10.00)	-1.510** (-2.16)
N	406	360
R-sq	0.918	0.914
F-test		
(Prob > F)	0.0000	0.0000

t statistics in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Source: Author's calculation

Table 2. Fixed effects model results

	UMIs	LMIs
ln_FDI	0.0628*** (4.09)	0.0684*** (4.54)
GDP	0.00935*** (2.64)	-0.000788 (-0.17)
GOV_EX	0.00567** (2.39)	-0.00161 (-1.41)
ln_EX_RATE	-0.217*** (-4.78)	-0.252*** (-7.09)
ln_POP	-0.461*** (-2.92)	0.887*** (3.61)
ln_INF	0.0199 (1.20)	-0.0326* (-1.73)
ln_TRADE	14.79*** (27.11)	11.05*** (15.89)
HDI	0.179* (1.94)	0.785*** (11.78)
Constant	18.92*** (7.73)	-2.267 (-0.57)
N	406	360
R-sq	0.918	0.914
F-test (Prob > F)	0.0000	0.0000

t statistics in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Source: Author's calculation

Appendix C

Code used for Table 3:

```
xtset COUNTRY_ID YEAR
sum
```

Code used for Table 4:

```
pwcorr, star(95)
```

Code used for Table 5:

```
reg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI ln_TRADE
vif
```

Code used for Table 6:

** REG model:*

```
reg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI ln_TRADE  
est sto reg
```

** FEM model:*

```
xtreg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI  
ln_TRADE, fe  
est sto fe
```

** REM model:*

```
xtreg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI  
ln_TRADE, re  
est sto re
```

** Hausman test:*

```
hausman fe re
```

** Breusch-Pagan test:*

```
reg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI ln_TRADE  
xttest3
```

** Wooldridge test:*

```
reg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI ln_TRADE  
xtserial ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI  
ln_TRADE
```

** Summary result:*

```
esttab reg fe re, r2 star(* 0.1 ** 0.05 *** 0.01) nogap compress
```

Code used for Table 7:

**47 countries in Asia:*

```
xtabond2 ln_EXP l.ln_EXP ln_FDI GDP GOV_EX ln_EX_RATE ln_POP ln_INF  
ln_TRADE HDI, gmm(L3.ln_EXP L.ln_FDI , lag(1 2) collapse) iv(GDP L2.GOV_EX  
ln_EX_RATE ln_POP ln_INF ln_TRADE HDI) two robust  
est sto xta
```

**UMIs:*

```
xtabond2 ln_EXP l.ln_EXP ln_FDI GDP GOV_EX ln_EX_RATE ln_POP ln_INF  
ln_TRADE HDI if UMI==1, gmm(L3.ln_EXP L.ln_FDI , lag(6 6) collapse) iv(GDP  
L2.GOV_EX ln_EX_RATE ln_POP ln_INF ln_TRADE HDI) two robust  
est sto xta1
```

**LMIs:*

```
xtabond2 ln_EXP l.ln_EXP ln_FDI GDP GOV_EX ln_EX_RATE ln_POP ln_INF  
ln_TRADE HDI if LMI==1, gmm(L3.ln_EXP L.ln_FDI , lag(2 4) collapse) iv(GDP  
L2.GOV_EX ln_EX_RATE ln_POP ln_INF ln_TRADE HDI) two robust  
est sto xta2
```

** Summary result:*

```
esttab xta xta1 xta2, r2 star(* 0.1 ** 0.05 *** 0.01) nogap compress
```

Code used for Table 1 (Appendix B):

**UMIs:*

```
reg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI ln_TRADE  
UMI==1
```

```
est sto reg1
```

**LMIs:*

```
reg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI ln_TRADE  
LMI==1
```

```
est sto reg2
```

** Summary result:*

```
esttab reg1 reg2, r2 star(* 0.1 ** 0.05 *** 0.01) nogap compress
```

Code used for Table 2 (Appendix B):

**UMIs:*

```
xtreg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI  
ln_TRADE UMI==1, fe
```

```
est sto fe1
```

**LMIs:*

```
xtreg ln_EXP ln_FDI GDP1 GOV_EX1 ln_EX_RATE ln_POP1 ln_INF HDI  
ln_TRADE LMI==1, fe
```

```
est sto fe2
```

** Summary result:*

```
esttab fe1 fe2, r2 star(* 0.1 ** 0.05 *** 0.01) nogap compress
```

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